

Problem Statement: Development of a Platform for Detection of Illicit Drug Sales on Darknet and Other Encrypted Platforms

Explanation: The increasing use of darknet marketplaces and encrypted online platforms has enabled organized criminal networks to facilitate the illegal sale and distribution of narcotic drugs while attempting to conceal their identities and activities. The anonymous and dynamic nature of these platforms presents significant challenges for law enforcement agencies in identifying suspicious activities, tracking criminal networks, and generating timely intelligence. Existing investigative approaches often require extensive manual effort and fragmented analysis across multiple digital sources, highlighting the need for a comprehensive intelligence platform capable of supporting efficient, data-driven investigations while adhering to legal and ethical frameworks.

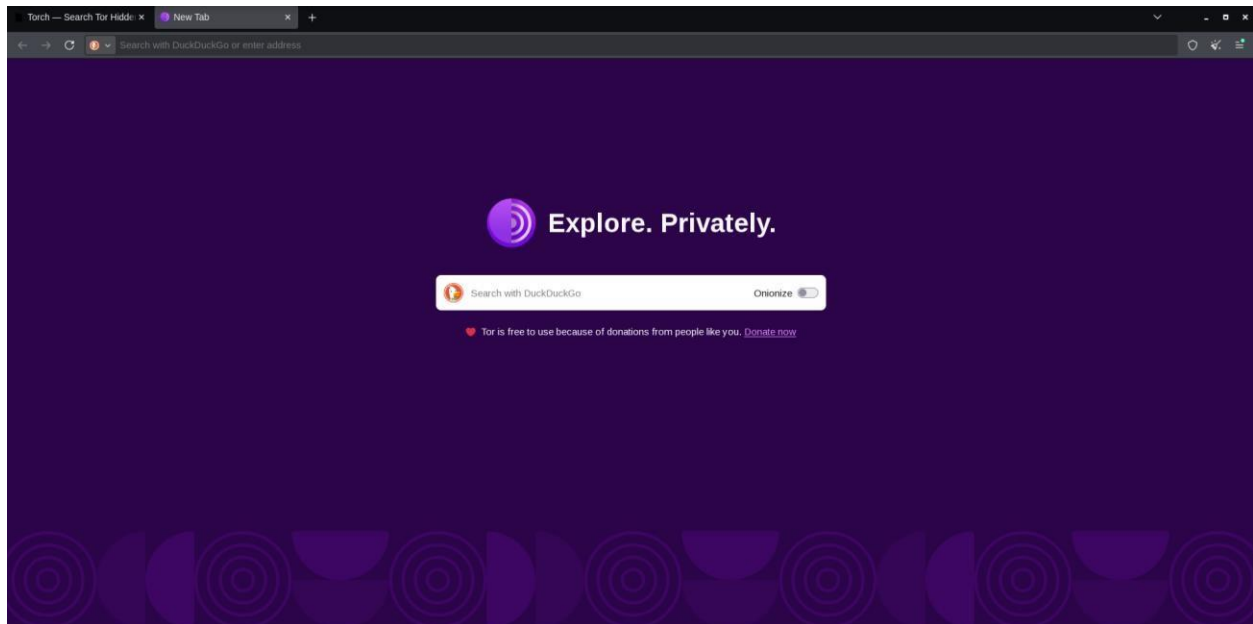


Figure 1: Tor Browser interface illustrating access to onion services (The screenshot is provided for educational purposes to demonstrate the type of environment where anonymity-focused services operate)

Objectives

- To assist law enforcement agencies in the early detection of illicit drug trafficking activities in digital ecosystems.

- To enhance cyber intelligence by integrating and correlating information from multiple lawful digital sources.
- To support investigators in identifying suspicious patterns, entities, and potential trafficking networks.
- To improve the speed, accuracy, and efficiency of digital investigations through automated intelligence generation.
- To strengthen evidence-based decision-making while adhering to applicable legal, ethical, and privacy frameworks.

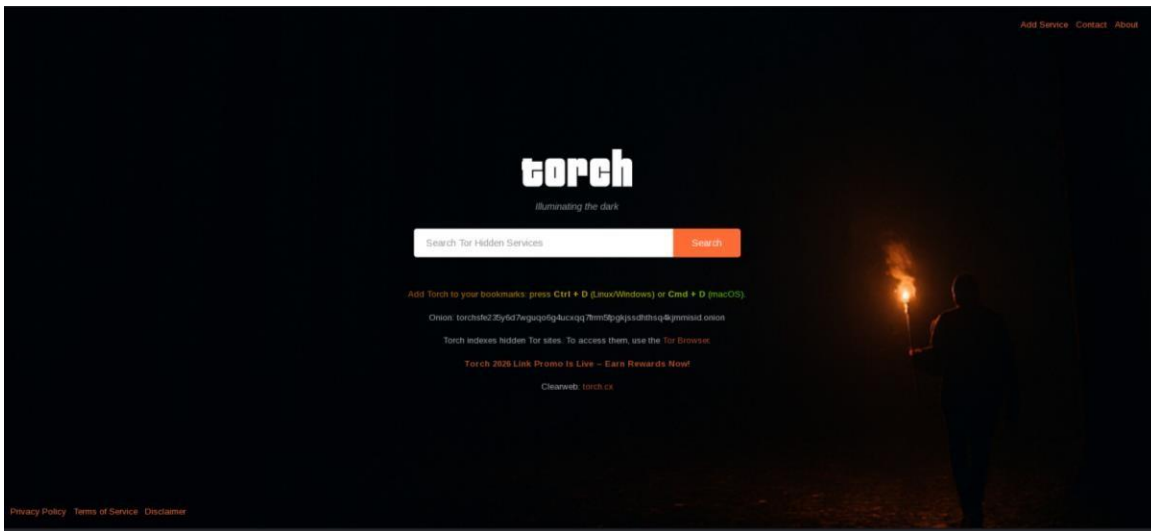


Figure 2: Torch Search Engine Interface (Darknet Search Engine). (The screenshot is provided solely for educational and research purposes to demonstrate the type of environment where anonymity-focused services operate. The image does not endorse or promote any illegal activity)

Expected Features

The proposed platform may include the following capabilities:

1. Multi-Source Data Collection

Enable collection and aggregation of information from legally accessible darknet marketplaces, public forums, blockchain intelligence sources, and other relevant digital platforms.

2. Intelligent Entity Correlation

Identify relationships between usernames, cryptocurrency wallets, email addresses, communication channels, and other digital identifiers to assist investigators in discovering potential criminal networks.

3. Suspicious Activity Detection

Automatically detect unusual patterns, recurring keywords, high-risk listings, or abnormal behavioral trends that may indicate illicit drug trafficking activities.

4. Interactive Intelligence Dashboard

Provide investigators with a centralized dashboard to monitor alerts, visualize trends, search intelligence records, and manage ongoing investigations.

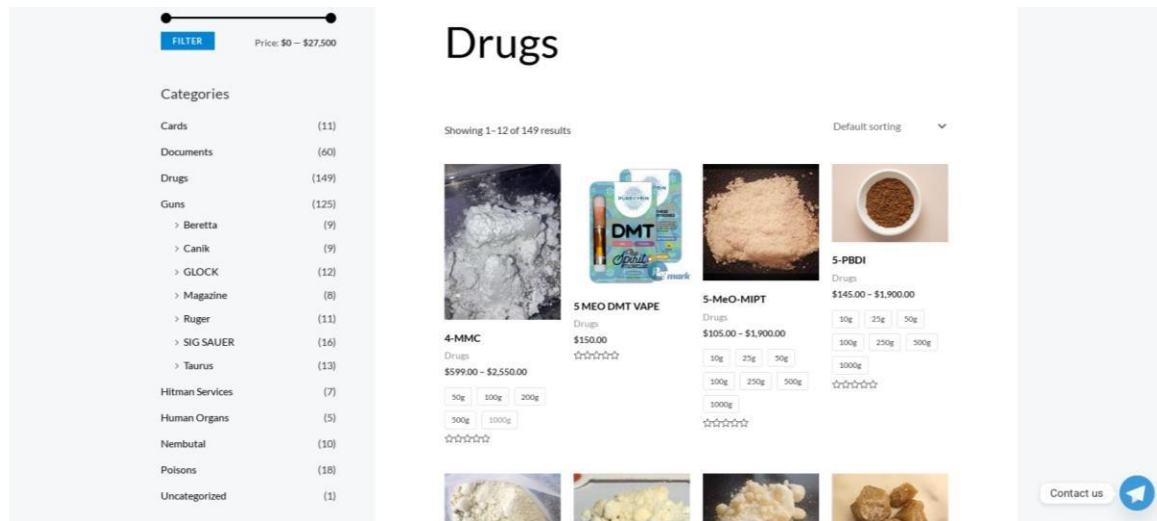


Figure 3. Representative darknet marketplace interface displaying illicit product listings. The screenshot is provided solely for educational and research purposes to illustrate the type of environment exploited for illegal online drug trafficking. It does not endorse, promote, or facilitate any unlawful activity.

5. Network Visualization

Display graphical relationships between suspects, marketplaces, cryptocurrency transactions, and communication channels to simplify complex investigations.

6. Automated Alert Generation

Notify investigators whenever predefined risk indicators or suspicious activities are detected across monitored sources.

7. Search & Investigation Support

Allow investigators to perform advanced searches using keywords, aliases, wallet addresses, or other identifiers while maintaining historical records for future reference.

8. Reporting & Evidence Management

Generate structured intelligence reports, maintain investigation logs, and facilitate secure documentation of digital evidence.

9. Security & Access Control

Ensure secure authentication, role-based access, audit logging, and protection of sensitive investigative information.

10. Scalability

Support continuous monitoring and analysis of large volumes of heterogeneous digital information with the ability to integrate additional intelligence sources in the future.

Tech Stack:

Participants are encouraged to justify their technology choices based on scalability, security, maintainability, performance, and suitability for the proposed solution rather than adopting any predefined technology stack.

References:

- 1) Real-World Operation: [Link](#)**
- 2) Official Report: [Link](#)**
- 3) Crime Report: [Link](#)**